CCE-2

Unitno:3 - Linked List.

Title: Train Coach Management System using stack and Queue in c++.

Problem Statement:

Design and implement a console-based Train Coach Management System using c++ and data structure. The system represents train coaches and allows coaches to be added or removed. A stack is used to add and remove coaches according to the Last In, first Out (LIFO) Principle, while a queue is used to manage coaches waiting to be attached to the train according to the first In, first Out (FIFO) principle. The system also displays the current coaches and provides a menu for performing different operations.

Objective:

- To implement a Train Coach Management System using c++.
- To understand the working of stack and queue data structures.
- To add and remove train coaches using queue stack operations.
- To manage waiting coaches using queue operation.

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- To display the current coaches & waiting coaches.
- To understand the practical applications of LIFO and FIFO in railway coach management.

Algorithm:

1. Start the program.

2. Initialize an empty stack for attached train coaches.

3. Initialize an empty queue for waiting coaches.

4. Display the menu of operations.

5. Accept the user's choice.

6. If the user selects AddCoach, enter the coach number and type, then push the coach onto the stack.

7. If the user selects RemoveCoach, remove the top coach using the pop() operation. Display a message if the stack is empty.

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8. If the user select Add Waiting Coach, enter the coach details and insert the coach into the queue using enqueue().

9. If the user selects Attach Waiting Coach, remove the first wating coach using dequeue() and attach it to the train stack.

10. If the user select Display Coaches, display all attached coaches and waiting coaches.

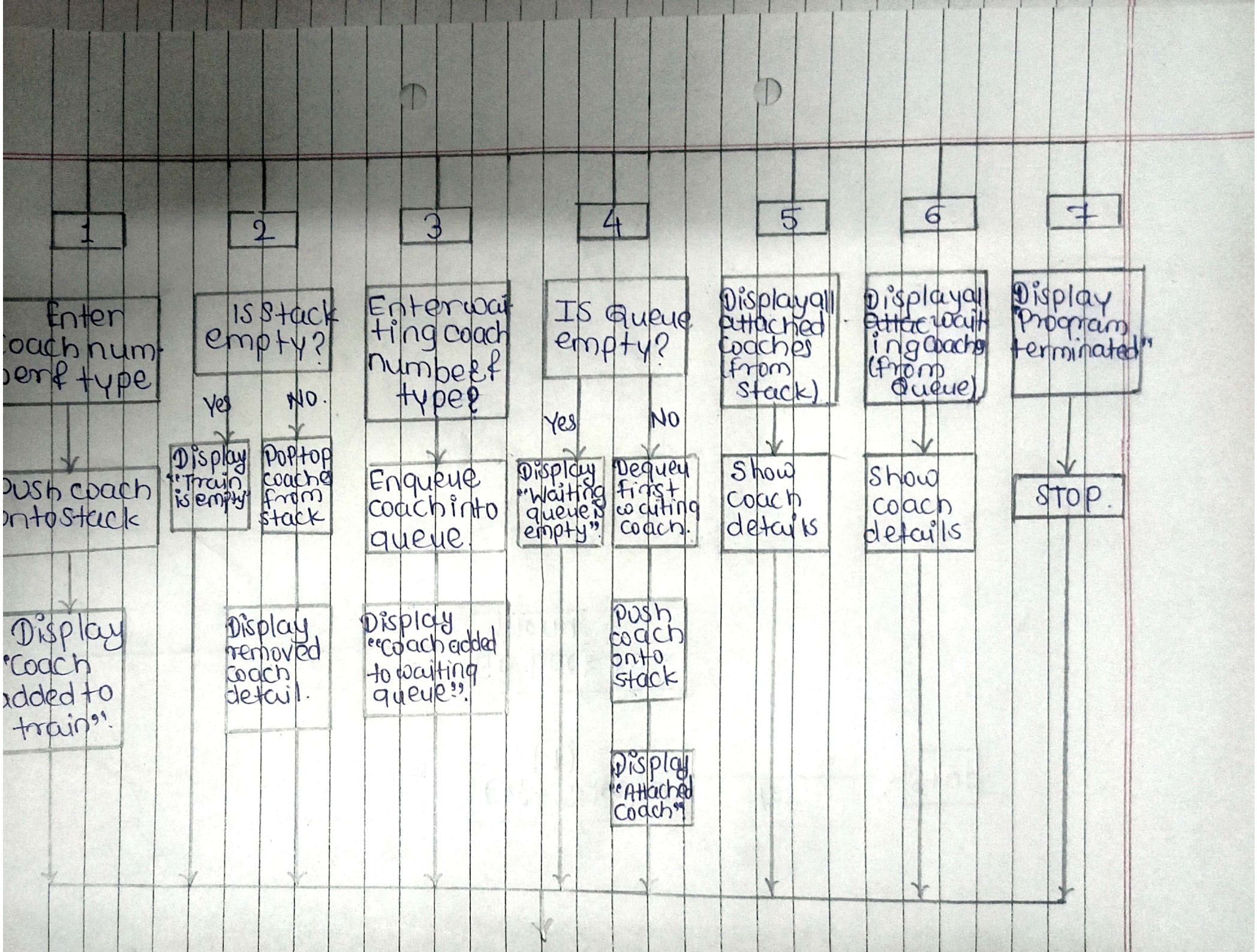
11 Repeat the menu until the user select Exit.

12. Stop the program.

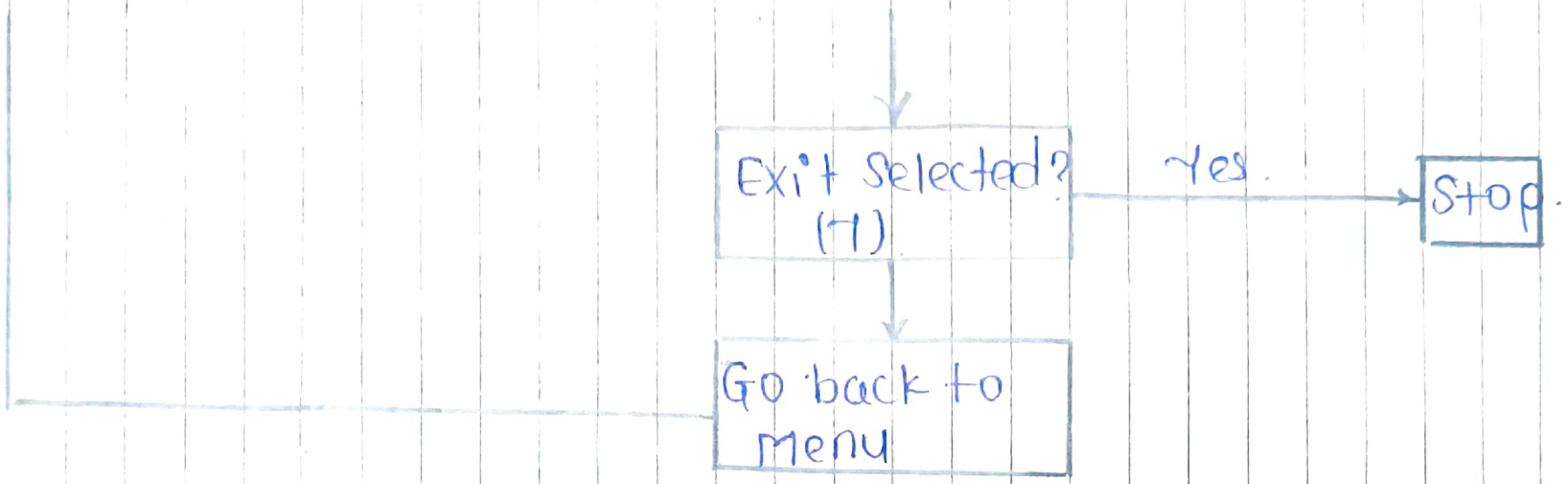
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Train Coach Management (using Stack & Queue)

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